

Understanding Tree Model Dominance in Tabular Learning - Performance, Robustness, and Stability Across Classical, Deep and Foundation (pre-trained) Models

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Abstract—Tree-based models remain highly competitive on tabular learning tasks despite the rapid progress of deep learning. In this work, we present a systematic comparison of tree-based, neural, and recent pretrained tabular models across a diverse benchmark of classification and regression datasets. Our study goes beyond clean-data evaluation by combining three perspectives: predictive performance after fair hyperparameter optimization, robustness under controlled data challenges, and stability under covariate shift. We evaluate classical tree ensembles (Random Forest, LightGBM, and CatBoost), neural baselines (MLP and TabNet), and recent tabular foundation-style models (TabPFN and APT) on six datasets, including two regression variants of the Diamonds dataset with different correlation-filtering thresholds. To assess robustness, we introduce missingness and synthetic high-cardinality categorical features, and we analyze clean versus shifted test distributions using both predictive degradation and SHAP-based stability measures. Across datasets and evaluation settings, tree-based methods consistently deliver the strongest and most reliable results, particularly under perturbation and shift. Deep models remain competitive in selected cases, but are generally more sensitive to feature-space changes, hyperparameter choice, and attribution instability. Pretrained tabular models reduce part of this gap, yet they do not consistently surpass boosted tree methods. These findings suggest that the continued strength of tree-based models arises not only from raw predictive accuracy, but also from their robustness and stability in realistic tabular settings.

Index Terms—tabular machine learning, tree-based models, deep learning, hyperparameter optimization, robustness, covariate shift, SHAP, feature attribution stability

I. INTRODUCTION

I-A. Background and Motivation

Tabular data is one of the most common data formats in applied machine learning and appears in domains such as finance, healthcare, logistics, and industrial analytics. Although deep learning has achieved major success in computer vision and natural language processing, tree-based methods still remain the strongest baseline for many tabular prediction tasks. In practice, models such as Random Forest, XGBoost, LightGBM, and CatBoost are often difficult to outperform, especially on datasets with mixed feature types, moderate sample sizes, and irregular data structure.

This continued strength raises an important question: why do tree-based models remain so effective on typical tabular data? A comparison based only on clean benchmark accuracy is not sufficient to answer this. In realistic settings, models must also remain reliable under incomplete inputs, difficult categorical structure, and changes in test-time data distribution. Understanding tabular model quality therefore requires not only predictive benchmarking, but also robustness and stability analysis.

II. RELATED WORK

Tree-based ensembles remain the dominant classical baseline for supervised tabular learning. XGBoost established a scalable and highly effective gradient-boosting framework, while CatBoost further strengthened tabular performance through ordered boosting and explicit handling of categorical features [1], [2]. These models remain central reference points because they combine strong predictive accuracy with practical robustness and efficient training on many structured datasets.

A large body of work has attempted to close this gap with neural architectures specialized for tabular data. TabNet introduced sequential attentive feature selection as an interpretable deep alternative [3], while TabTransformer and SAINT extended this direction through attention-based contextualization and contrastive pretraining [4], [5]. Gorishniy et al. later showed that carefully tuned simple baselines, especially ResNet-style models and FT-Transformer, are substantially stronger than many earlier bespoke tabular neural networks, but still do not consistently displace boosted tree ensembles [6].

Recent benchmark efforts have broadened this discussion. McElfresh et al. introduced TabZilla, a large-scale benchmark showing that the relative performance of neural and tree-based methods depends strongly on dataset characteristics and evaluation design [7]. Erickson et al. introduced *TabArena*, a continuously maintained living benchmark for tabular machine learning, emphasizing protocol-aware evaluation and public leaderboards [8]. The study most directly related to our interpretation is that of Grinsztajn et al., who argue that tree-based models remain strong on typical tabular data because

their inductive biases are better aligned with uninformative features, feature orientation, and irregular decision structure [9].

More recently, pretrained and zero-shot tabular models have shifted part of the comparison. TabPFN frames tabular prediction as in-context inference using a transformer pretrained on synthetic tasks, giving strong performance especially on small classification problems [10]. APT extends this direction through adversarially pretrained tabular prediction [11]. These models reduce part of the tuning burden, but their robustness and feature-stability behavior under perturbation and shift remain less established than their clean benchmark performance.

Robustness-oriented tabular evaluation has also received increasing attention. TableShift provides a benchmark for distribution shift in tabular data across several real-world domains, showing that in-distribution and out-of-distribution performance can diverge substantially [12]. Gardner et al. further show that tree-based methods are strong baselines for subgroup robustness, often competing with or outperforming specialized robustness and fairness methods [13]. Tabular-Bench studies adversarial robustness for tabular deep learning, highlighting that robustness in structured data requires different assumptions from image-based adversarial settings [14]. Compared with these works, our study focuses on a complementary setting: controlled missingness and high-cardinality stress, covariate-shift evaluation, and SHAP-based feature-stability analysis within one tree-versus-neural-versus-pretrained comparison.

III. EXPERIMENTAL SETUP

III-A. Datasets

We selected six tabular datasets spanning both classification and regression tasks in order to evaluate model behavior across different feature structures, target types, and levels of learning difficulty. The classification benchmark consists of the *Titanic*, *Telco Churn*, *DryBean*, and tabularized *MNIST* datasets, while the regression benchmark consists of two curated variants of the Diamonds dataset: *diamonds_v2* and *diamonds_v3*.

The datasets were chosen deliberately to provide diversity in sample size, dimensionality, feature composition, and target structure. In particular, the benchmark includes binary and multiclass classification, purely numerical as well as mixed numerical–categorical inputs, and regression settings with strong feature–target dependence. This diversity is important because the relative strengths of tree-based, neural, and pretrained tabular models are often shaped by properties such as feature heterogeneity, redundancy, categorical complexity, and dataset scale.

Each dataset also plays a distinct structural role in the benchmark. *Titanic* represents a small mixed-feature binary classification problem in which limited sample size and heterogeneous attributes can favor models with strong inductive bias. *Telco Churn* provides a medium-scale binary classification task with many categorical predictors, making it particularly relevant for studying categorical handling and sensitivity to nuisance structure. *DryBean* is a multiclass benchmark with primarily numerical shape descriptors, allowing us to assess

TABLE I: Summary of datasets used in this study.

Dataset	Task	Samples	Feat.	Num/Cat	Note
diamonds_v2	Reg.	219,703	24	6/18	Pearson 0.85
diamonds_v3	Reg.	219,703	22	4/18	Pearson 0.70
drybean	Multi-cl.	4,970	12	12/0	Bean benchmark
mnist	Multi-cl.	1,797	64	64/0	Tabular digits
telco	Binary cls.	7,032	20	4/16	Churn dataset
titanic	Binary cls.	183	11	6/5	Survival task

model behavior in a structured geometric setting. Tabularized *MNIST* serves as a high-dimensional numerical classification task, where the advantages of deep models may become more pronounced than in smaller mixed-type datasets (The file used in this project is not the original 70k-sample MNIST dataset. It is the full 8×8 tabular digits benchmark, which naturally contains 1,797 samples and 64 features).

The two Diamonds variants were included to study regression behavior under different levels of feature redundancy control. Both are derived from the same source dataset, but they differ in the correlation threshold used during feature filtering. In *diamonds_v2*, features were filtered using a Pearson correlation threshold of 0.85, whereas *diamonds_v3* used a stricter threshold of 0.70, removing more moderately correlated predictors. This distinction allows us to assess whether stronger decorrelation affects model behavior differently, particularly for neural models that may respond more sensitively to changes in feature redundancy.

A summary of the datasets, including task type, number of samples, and feature composition, is provided in Table I.

III-B. Models

We evaluated three groups of tabular learning models: *tree-based methods*, *neural tabular models*, and *pretrained tabular foundation-style models*.

The tree-based group includes *Random Forest (RF)*, *XGBoost*, *LightGBM*, and *CatBoost*. These models were selected because they represent the strongest and most widely used classical approaches for tabular prediction. Random Forest serves as a robust bagging-based ensemble baseline, while XGBoost, LightGBM, and CatBoost provide modern gradient-boosted tree frameworks with different design choices for efficiency, split construction, and categorical feature handling.

The neural group includes a standard *multilayer perceptron (MLP)* and *TabNet*. The MLP was included as a simple dense neural baseline, whereas TabNet was chosen as a tabular-specific neural architecture based on sequential attention and feature selection. These two models allow comparison between a general-purpose neural approach and a specialized deep tabular model.

In addition, we included two recent pretrained tabular models, *TabPFN* and *APT*. Unlike the other models, these methods are designed for zero-shot or low-tuning inference using pretrained representations rather than extensive dataset-specific optimization. They were therefore evaluated using their default pretrained configurations instead of the full hyperparameter optimization procedure. Their inclusion broadens the benchmark beyond conventional neural baselines and allows us to assess whether recent pretrained tabular models narrow the gap to tree-based methods.

A compact summary of the evaluated models is given in Table II.

TABLE II: Summary of evaluated model families and tuning strategy.

Model	Family	Task	HPO	Note
RF	Tree	Cls./Reg.	Yes	Bagging ensemble
XGBoost	Tree	Cls./Reg.	Yes	Gradient boosting
LightGBM	Tree	Cls./Reg.	Yes	Histogram boosting
CatBoost	Tree	Cls./Reg.	Yes	Native categorical support
MLP	Neural	Cls./Reg.	Yes	Dense feed-forward network
TabNet	Neural	Cls./Reg.	Yes	Sequential attention model
TabPFN	Pretrained	Cls.	No	Default pretrained config
APT	Pretrained	Cls.	No	Fixed pretrained checkpoint

III-C. Hyperparameter Optimization Protocol

To ensure a fair comparison, we applied hyperparameter optimization (HPO) separately for each dataset–model pair using Optuna. For the tunable models, each optimization run used 30 trials, and the best configuration was selected according to validation performance before final evaluation on the held-out test set.

Optuna was chosen because it provides an efficient and flexible search framework for heterogeneous model families with mixed continuous, discrete, and categorical hyperparameters. Compared with manual tuning or fixed grid search, this approach allows broader exploration of relevant parameter combinations while avoiding the inefficiency of exhaustively evaluating implausible settings. This is particularly important in a benchmark containing both tree ensembles and neural models, whose parameter spaces differ substantially in scale and structure.

For tree-based models, the search spaces included parameters such as the number of trees, maximum depth, learning rate, subsampling, column sampling, and regularization terms, depending on the model family. For neural models, we tuned architectural and optimization parameters including the number of layers, hidden dimensions, dropout, learning rate, batch size, and, for TabNet, architecture-specific parameters such as decision and attention dimensions.

III-D. Robustness stress scenarios

These perturbations were applied to four datasets—*diamonds_v2*, *Telco*, *DryBean*, and tabularized *MNIST*—selected to cover regression, binary and multiclass classification, and both mixed-type and purely numerical feature settings. *Titanic* was excluded because it already contains substantial intrinsic missingness and has a small sample size, making additional synthetic stress less clean to interpret. *diamonds_v3* was excluded because it is a structural variant of *diamonds_v2*, so including both would introduce unnecessary duplication at this stage.

For the missingness scenario, we randomly removed 15% of feature values from 20% of randomly selected columns in both the training and test sets. The goal was to simulate moderate but nontrivial incompleteness under a controlled and reproducible perturbation level. Missing values were then handled through each model’s preprocessing pipeline. Numerical

features were imputed using a central-tendency rule, while categorical features were assigned either the most frequent level or a dedicated “Missing” category, depending on the model family and preprocessing design. After this transformation, each model was retrained using the same hyperparameter configuration selected during the clean-data HPO stage and then evaluated on the correspondingly perturbed test set.

For the high-cardinality scenario, we introduced two synthetic categorical variables, each containing 150 unique categories, into both the training and test data. These variables were generated at random and were not intended to carry predictive signal. Their purpose was to simulate nuisance categorical structure of the kind frequently encountered in tabular applications, such as ID-like or highly sparse categorical fields. The perturbed datasets were then processed using the encoding strategy appropriate to each model family. Tree-based methods received ordinal-style categorical representations, while neural models used the corresponding expanded representation required by their preprocessing pipeline. As in the missingness experiment, all models were retrained on the augmented data using the best clean-data hyperparameters rather than retuned specifically for the perturbation.

For each robustness scenario, we recorded predictive performance, training time, and model size. For classification tasks, robustness was assessed using Accuracy and F1-score; for regression tasks, we report R^2 , RMSE, and MAE where appropriate. By comparing the clean, missingness, and high-cardinality settings, we can quantify not only absolute predictive degradation but also whether different model families remain computationally stable when the feature space becomes incomplete or structurally more difficult.

III-E. Covariate shift and SHAP stability protocol

Beyond explicit robustness stress tests, we also investigated model behavior under covariate shift. This analysis was conducted on the three classification datasets—*Titanic*, *Telco*, and *DryBean*—because they together cover binary and multiclass classification, mixed feature structures, and semantically interpretable predictors, making SHAP-based stability comparisons more meaningful. We did not include the regression datasets because this stage was designed around joint evaluation of Accuracy/F1 degradation and feature-attribution stability in classification settings. Tabularized *MNIST* was excluded because its image-derived feature space is less aligned with the feature-semantics questions emphasized in the present shift analysis.

For each dataset, models were trained on the original training data and then evaluated on two versions of the test set: the original clean test set and a shifted test set constructed through controlled perturbations of both numerical and categorical features. For numerical variables, the shifted test set was created by applying a small mean shift, additive Gaussian noise, and a small fraction of outlier injections. Concretely, numerical features received a mean shift of 0.2σ , additive Gaussian noise with standard deviation 0.5σ , and 2% outlier perturbations of magnitude 3σ , where σ denotes the feature-specific standard deviation. For categorical variables, 10% of

values were replaced with the most frequent category and an additional 5% were replaced with a previously unseen category. This protocol was intended to simulate moderate distributional drift without changing the underlying target definition.

IV. HYPERPARAMETER OPTIMIZATION RESULTS

IV-A. Cross-dataset performance comparison

We begin by comparing model performance after hyperparameter optimization across the full benchmark. Because the study includes both classification and regression tasks, raw metric values are not directly comparable across datasets. We therefore aggregate results using within-dataset ranks together with summary statistics such as wins, top-2 frequency, and average gap to the best model.

A clear pattern emerges from Table III the strongest boosted tree methods occupy the most favorable region of the benchmark. XGBoost, CatBoost, and LightGBM all remain near the top in overall rank, and their average gaps to the best model stay small. This is important because it indicates not only occasional wins, but repeated near-best placement across heterogeneous tasks. In other words, their advantage is not driven by one unusually favorable dataset or one isolated hyperparameter outcome.

The three leading boosted tree models all achieve overall mean ranks in the narrow range of roughly 2.2 to 2.7, with average gaps to the best model remaining below about 6%.

The neural models show a different profile. They can be competitive on selected datasets, but their overall placement is less stable and their average gaps are wider. This suggests that neural performance is more localized: strong results are achievable, but less consistently across datasets and less robustly across configurations. The practical implication is that the tree-based advantage is not only about peak score, but also about repeatability across the benchmark.

The pretrained models provide a useful contrast. TabPFN reduces part of the gap to the tuned neural baselines, especially on some classification tasks, but it does not approach the cross-dataset consistency of the strongest boosted trees. APT is clearly weaker in this benchmark. Taken together, the results show that the strongest tree-based models are not merely the most accurate on average; they are also the most consistently competitive across diverse tabular settings.

IV-B. Classification and regression performance patterns

Across the classification datasets, tree-based methods remain the most consistently competitive family. Their advantage is especially visible on smaller or mixed-type problems, where strong inductive bias and selective use of informative predictors appear to matter more than representational flexibility alone. On such tasks, the strongest tree models do not merely win by a narrow margin; they frequently maintain a clear buffer over the neural baselines, indicating that the difficulty is not simply insufficient tuning of deep models, but a deeper mismatch between model inductive bias and the structure of the tabular task.

At the same time, the classification results are not uniform across all datasets, and this variation is informative. On tasks that are more homogeneous or more strongly numerical, the gap between trees and neural models narrows. This suggests that the tree advantage is not absolute, but depends on the extent to which the dataset rewards selective splitting on stable predictors rather than high-capacity distributed representations. In other words, the benchmark does not support the simplistic claim that neural models always underperform on tabular data; instead, it suggests that they become more competitive when the input structure is more aligned with the strengths of dense representation learning.

The regression results reinforce this interpretation while adding an important structural contrast through *diamonds_v2* and *diamonds_v3*. Because both variants are derived from the same source dataset but differ in correlation-filtering threshold, they allow us to observe how stronger redundancy control affects different model families. The benchmark indicates that tree-based methods remain comparatively stable under this change, whereas neural models are more sensitive to the removal of moderately correlated predictors.

This Diamonds contrast is particularly relevant given the broader motivation of the paper. One of the recurring questions in tabular learning is whether tree-based models benefit from their ability to preserve feature orientation and exploit informative thresholds without depending heavily on distributed representations. The relative stability of tree models across *diamonds_v2* and *diamonds_v3* is consistent with that view. Neural models, by contrast, may benefit more from feature redundancy or moderately correlated support variables, and therefore exhibit greater sensitivity when that redundancy is filtered more aggressively.

IV-C. Predictive quality versus computational and model-size trade-offs

Predictive performance alone does not fully determine model suitability for tabular learning. Figure 1 therefore summarizes the average training time and model size across datasets.

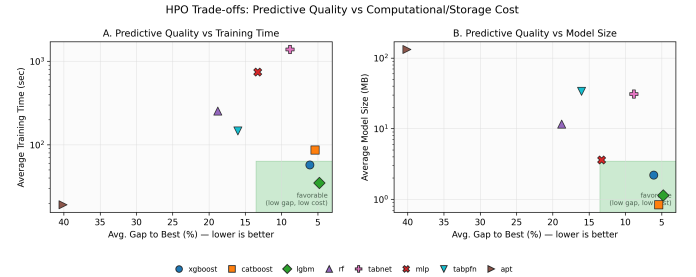


Fig. 1: Predictive quality versus computational and storage cost after hyperparameter optimization. Panel A compares average gap to the best model with training time, and Panel B compares average gap to the best model with model size. Models closer to the lower-right region combine stronger predictive quality with lower cost.

The results indicate that stronger performance does not always require disproportionate computational cost, but neither

Model	Mean Rank (Cls.)	Mean Rank (Reg.)	Mean Rank (Overall)	Wins	Top-2	Avg. Gap to Best (%)	Time Rank	Size Rank
xgboost	2.25	2.00	2.17	1	4	6.09	2.83	3.50
catboost	2.00	3.50	2.50	2	3	5.40	4.67	2.00
lgbm	1.75	4.50	2.67	2	3	4.80	3.00	2.83
rf	4.00	1.00	3.00	3	3	18.78	3.67	4.83
tabnet	3.50	5.50	4.17	2	2	8.83	7.00	5.00
mlp	6.00	4.50	5.50	0	0	13.29	5.17	3.00
tabpfn	5.75	—	5.75	0	0	16.04	5.50	6.75
apt	7.75	—	7.75	0	0	40.11	1.50	8.00

TABLE III: HPO model summary across datasets. Ranks are computed within each dataset first, then averaged across datasets. Lower is better for rank/gap/time/size; higher is better for Wins and Top-2. Avg. Gap to Best (%) is computed per dataset and then averaged per model. For each dataset, we take the best score among all models (F1 for classification, R^2 for regression) as the reference, compute each model’s percentage shortfall from that reference, and then average these gaps across datasets. Thus, lower values indicate better predictive quality, and 0% means the model matches the dataset-best score.

is there a simple monotonic relation between accuracy and efficiency.

A practically important pattern is that the strongest tree-based methods often sit near a favorable performance–efficiency frontier. Their predictive advantage is therefore reinforced by operational convenience: they are not only strong in score, but also competitive in training cost and storage requirements. Neural models, in contrast, may require greater computational burden or larger effective representations while still exhibiting less stable benchmark-wide ranking. This does not make them uncompetitive, but it does reduce their attractiveness when predictive gains are small or inconsistent. The pretrained models occupy an intermediate position: they reduce dataset-specific tuning burden, but this convenience does not automatically translate into superior overall predictive or efficiency trade-offs.

IV-D. Statistical ranking across classification datasets

To complement the raw classification results, we compared model ranks across the four classification datasets using a Friedman test followed by Nemenyi post-hoc analysis at $\alpha = 0.05$. The resulting critical-difference diagram should be interpreted cautiously: its main message is not a strict total ordering of models, but the extent to which observed rank differences are statistically distinguishable.

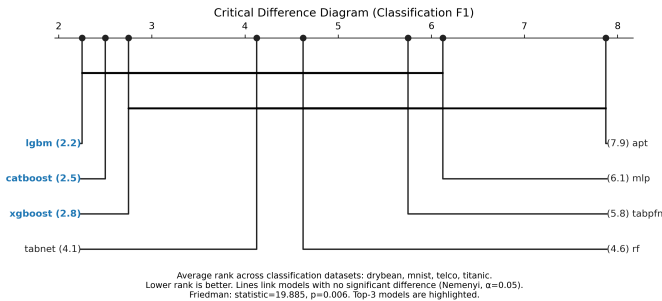


Fig. 2: Critical difference diagram for classification datasets based on F1-score ranks.

In this benchmark, several pairwise differences are not statistically significant, even when the mean-rank gaps appear

visually large. This is plausible for two reasons. First, the classification suite is small, containing only four datasets, which limits statistical power. Second, the Nemenyi procedure is conservative, especially when many models are compared simultaneously. In our case, the combination of eight models, a small number of datasets, and several tied or extreme outcomes increases the critical difference and makes broad non-significance unsurprising.

The diagram therefore supports a restrained conclusion. LightGBM and CatBoost occupy the strongest part of the ranking, while APT lies clearly on the weak end. Beyond those broad separations, however, many intermediate pairwise differences should not be over-interpreted. Thus, the rank-based analysis is most useful here as evidence that the strongest tree-based models remain concentrated near the top of the classification benchmark, rather than as proof of a finely resolved hierarchy among all methods.

V. ROBUSTNESS TO MISSINGNESS AND HIGH-CARDINALITY FEATURES

V-A. Robustness under missingness

The missingness results show that tree-based models remain the most stable overall when inputs are made incomplete. The key pattern in Table IV is not only that their average predictive degradation is small, but that this is accompanied by strong feature-retention behavior. CatBoost, LightGBM, and Random Forest all preserve high SHAP overlap under missingness, indicating that they continue to rely on nearly the same core predictors even after values are removed. This is an important distinction: missingness robustness is more convincing when performance retention and attribution stability move together rather than in isolation.

Among all evaluated models, CatBoost presents the strongest overall profile. Its mean predictive degradation remains close to zero, and its SHAP overlap stays near the top of the benchmark. LightGBM and Random Forest are similarly stable, though slightly weaker in attribution retention. By contrast, the neural models are less consistent. MLP shows a moderate predictive drop together with lower top-feature stability, suggesting that it remains more sensitive to disruption of the input space. TabNet is closer to the tree models in

TABLE IV: Model-level robustness summary under missingness and high-cardinality stress. Values are averaged over the available robustness datasets. Performance deltas are measured relative to the clean condition, so values closer to zero indicate greater predictive robustness; positive values indicate improvement under perturbation. Training-time change is reported as percentage change relative to clean training time. SHAP overlap is the average number of shared features in the top-10 feature sets between clean and perturbed conditions, and Jaccard is the corresponding top-10 set overlap ratio. XGBoost is excluded from this section because it was not part of the final robustness comparison. ‘NA’ denotes unavailable values, primarily because SHAP-based stability quantities were not computed for APT and TabPFN. Mean performance and training-time deltas are taken from the figure-level robustness summaries, while mean Jaccard values are averaged from the available SHAP-delta summary entries.

Model	Δ Perf. M	Δ Perf. H	Δ Time% M	Δ Time% H	Overlap M	Overlap H	Jaccard M	Jaccard H
CatBoost	-0.0119	-0.0108	-12.23	1.11	9.00	9.25	0.826	0.871
LightGBM	-0.0004	-0.0088	1.97	2.12	9.00	8.50	0.818	0.742
Random Forest	-0.0025	-0.0098	0.43	15.44	9.00	8.25	0.839	0.710
MLP	-0.0077	-0.0194	-7.88	493.18	7.00	5.00	0.617	0.361
TabNet	-0.0002	0.0225	-39.68	-46.62	6.25	5.50	0.537	0.422
APT	0.1200	0.0800	NA	NA	NA	NA	NA	NA
TabPFN	-0.0200	-0.3000	NA	NA	NA	NA	NA	NA

M = missingness, H = high-cardinality.

average predictive retention, but its feature-stability profile is weaker, indicating that comparable score retention does not necessarily imply equally stable internal reliance on features.

The efficiency statistics reinforce this interpretation. Under missingness, reduced training cost should not automatically be interpreted as robustness. In several cases, faster training appears to reflect a changed effective training regime rather than a genuinely stable response to incomplete inputs. Overall, the missingness experiment supports a conservative conclusion: tree-based models are not only less affected in final score, but also more likely to preserve the same explanatory structure under degraded inputs.

V-B. Robustness under high-cardinality categorical stress

The high-cardinality stress results produce the clearest separation between model families. Table IV shows that tree-based models, especially CatBoost, remain comparatively stable when irrelevant but structurally difficult categorical features are added. CatBoost exhibits the strongest combined profile: only a small predictive change, high SHAP overlap, and limited growth in model size or training burden. LightGBM and Random Forest also remain robust, although their feature-retention statistics are somewhat weaker than CatBoost’s.

The neural models are affected more strongly. The MLP shows the sharpest overall deterioration in this setting, not only through weaker feature-stability values but also through a dramatic increase in computational cost. This suggests that the synthetic categorical burden is not being ignored; instead, it expands the learned representation and makes optimization substantially harder. TabNet performs somewhat better than MLP in predictive terms, but its attribution stability remains clearly below that of the strongest tree models. In other words, the nuisance categorical structure does not merely reduce score for neural models; it also changes which predictors they appear to trust.

This stress setting therefore offers some of the strongest evidence in the paper for a structural advantage of tree-based methods. The tree ensembles appear better able to filter out irrelevant categorical complexity, whereas neural models are more easily displaced by it. That contrast is especially important because the added features are intentionally uninformative: a robust learner should ideally ignore them. On this criterion, the tree-based models are clearly stronger.

VI. FEATURE IMPORTANCE AND STABILITY UNDER COVARIATE SHIFT

Covariate shift probes a different form of robustness from the missingness and high-cardinality experiments. Here the issue is not incomplete inputs or added nuisance variables, but whether models trained on the original data remain stable when the test distribution changes while the task itself remains fixed. We evaluate this stability along two complementary axes: predictive retention and feature-importance retention. Table V reports the corresponding averages across the three shift datasets.

VI-A. Predictive stability under shift

The shift results show that clean-data strength and shifted-data stability are related, but not identical. CatBoost remains the strongest model in absolute terms, retaining the highest mean shifted Accuracy and F1, even though its average performance drop is not the smallest. This is an important distinction: smaller degradation does not automatically imply better shifted performance. Models such as MLP and APT show smaller average declines, but from clearly weaker clean baselines. In other words, a model can appear more stable in relative terms simply because it had less performance to lose.

The more meaningful interpretation is therefore joint rather than absolute. A strong shifted model should retain useful performance after shift, not merely minimize the drop from its

TABLE V: Average predictive and SHAP-based stability under covariate shift. Values are averaged over the three shift datasets. NA indicates unavailable values (e.g., models that failed in this environment, OpenMP SHM error while running model.predict, or models without SHAP stability computation)

Model	Acc. Clean	Acc. Shift	Δ Acc.	F1 Clean	F1 Shift	Δ F1	SHAP ρ	SHAP Jac.	$n_{\text{perf}} / n_{\text{SHAP}}$
CatBoost	0.932998	0.715998	-0.217000	0.864263	0.621144	-0.243119	0.992377	0.939394	3 / 3
RF	0.850254	0.704469	-0.145785	0.620262	0.474098	-0.146163	0.988502	0.878788	3 / 3
MLP	0.747757	0.676971	-0.070786	0.726536	0.636836	-0.089700	0.942586	0.698413	3 / 3
TabNet	0.818573	0.695785	-0.122788	0.843091	0.671167	-0.171924	0.947760	0.467172	3 / 3
TabPFN	0.798919	0.626937	-0.171982	0.787599	0.562084	-0.225515	NA	NA	3 / 0
APT	0.637568	0.593243	-0.044324	0.481393	0.402575	-0.078818	NA	NA	3 / 0
LGBM	NA	NA	NA	NA	NA	NA	NA	NA	0 / 0

Δ values are computed as shifted minus clean, so values closer to zero indicate greater predictive stability. SHAP ρ denotes mean Spearman rank correlation and SHAP Jac. denotes mean Jaccard overlap of the top-10 features. n_{perf} and n_{SHAP} indicate the number of datasets contributing to the predictive and SHAP aggregates, respectively.

clean score. On this criterion, CatBoost remains the clearest example of predictive robustness, while Random Forest also shows a comparatively stable profile. The broader implication is that covariate shift exposes a different kind of fragility from the robustness stresses in Section V: it tests not whether models can learn under degraded inputs, but whether they continue to generalize when the feature distribution itself is displaced.

VI-B. Feature-importance stability under shift

The SHAP analysis adds a second layer to this result by asking whether models continue to rely on the same dominant predictors after shift. On this criterion, the tree-based models are again the most stable. CatBoost achieves the strongest overall SHAP stability, and Random Forest follows closely, indicating that both models preserve not only the global ordering of feature importance but also most of the top-ranked feature set. This makes their shifted behavior more convincing: they are not merely retaining predictive quality, but doing so while remaining anchored to largely the same explanatory structure.

The neural models are less stable in this sense. MLP retains a moderate degree of feature-importance consistency, but TabNet is notably weaker in top-feature retention. This gap is especially informative because it shows that global stability and top-feature stability are not the same. A model may preserve a broadly similar ranking of feature importances while still changing which predictors it relies on most strongly. For interpretation, the latter is often more consequential.

A second important point is that SHAP stability should not be read as a guarantee of predictive robustness. A model can preserve its dominant predictors and still lose predictive quality if the shifted distribution weakens the usefulness of those predictors. Thus, predictive degradation and attribution drift are related but distinct. Taken together, the shift results strengthen one of the paper’s central conclusions: the strongest tree-based models are more reliable not only because they perform well after shift, but because they preserve a more stable notion of which features matter.

VII. CROSS-EXPERIMENT ANALYSIS

VII-A. Rank stability versus robustness

A central cross-experiment question is whether clean-data strength carries over to more demanding evaluation settings. The results suggest that it does, but only partially. The strongest clean benchmark models are also the ones most likely to remain competitive under missingness, nuisance categorical structure, and covariate shift. This indicates that clean-data rank is not arbitrary: it contains real information about broader model quality. At the same time, the relationship is not exact. Strong clean performance does not automatically imply equal robustness, and robustness itself depends on which kind of perturbation is introduced. This is precisely the relationship illustrated by Figure 3.

What matters most here is consistency across evaluation regimes. The strongest tree-based models are not simply top-ranked on the clean benchmark; they remain near the top when the evaluation moves from standard prediction to controlled stress and distributional displacement. This makes their advantage qualitatively different from isolated wins on individual datasets. It suggests that their strength is structural rather than accidental: the same model family that performs well under ordinary conditions also degrades more gracefully when the feature space becomes incomplete, cluttered, or shifted.

This perspective also helps clarify why some non-tree models appear promising but do not alter the broader conclusion. Neural and pretrained methods can reach competitive results in selected settings, yet their relative position is less stable once the benchmark is widened beyond clean accuracy alone. Thus, the main lesson of the combined experiments is not merely that tree-based models win more often, but that they remain more reliable when the evaluation criterion itself becomes stricter.

VII-B. Performance drop versus attribution drift

A second cross-experiment pattern is that predictive degradation and attribution drift are related, but clearly not interchangeable. Models that lose predictive quality under perturbation often also show weaker feature-stability behavior, yet the two effects do not move in lockstep. A model can preserve a broadly similar explanatory structure while still suffering a

Performance Retention vs Attribution Stability Under Perturbation

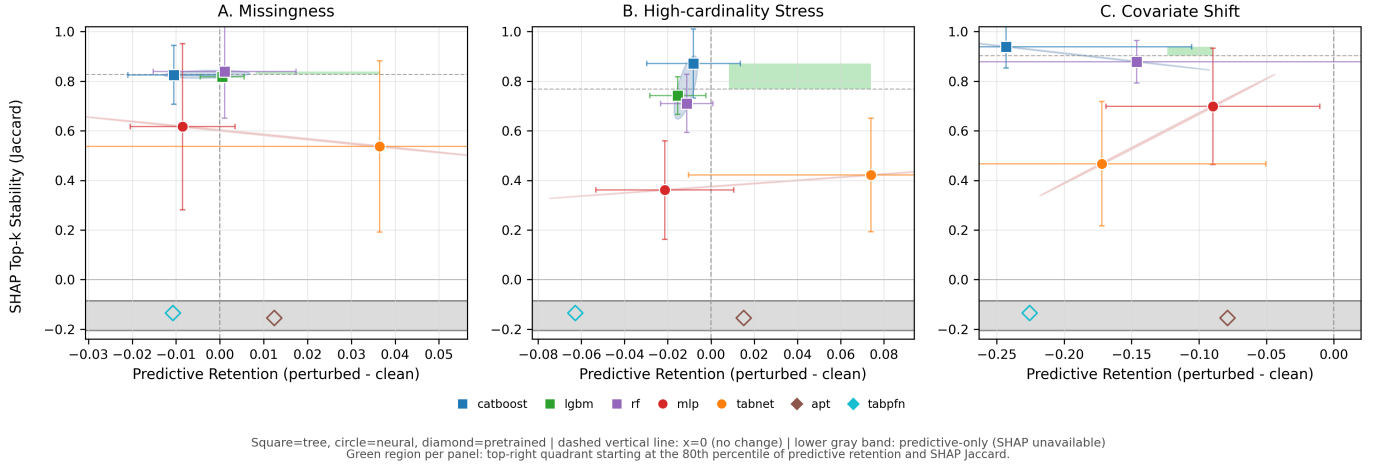


Fig. 3: Performance retention versus attribution stability under three perturbation types: missingness, high-cardinality stress, and covariate shift. Each point denotes a model. The x-axis reports predictive retention relative to the clean condition, and the y-axis reports SHAP top- k stability measured by Jaccard overlap. Error bars indicate dataset-level standard deviation where available. The lower shaded band marks models for which predictive results are available but SHAP-based stability values were not computed. Translucent regions indicate broad model-family groupings, stronger robustness corresponds to smaller predictive change (x -axis values closer to zero) together with higher attribution stability. Top-right green shaded quadrant starting at panel-wise 80th percentile of predictive retention and SHAP Jaccard

substantial performance drop, and it can also retain reasonable predictive quality while shifting which features it relies on most. For this reason, neither predictive metrics nor SHAP-based stability should be treated as a complete proxy for the other.

This distinction is especially important for interpretation. Predictive robustness tells us whether the model still performs well after the data changes. Attribution stability asks a different question: whether the model still appears to trust the same predictors. The latter matters because two models with similar shifted performance may differ strongly in how stable their internal reliance pattern remains. In practical terms, a model that preserves score only by reallocating importance across features is a weaker form of robustness than one that remains anchored to the same core signals.

The cross-experiment evidence therefore supports a more careful definition of robustness. A robust tabular model should not only resist predictive degradation, but should do so while preserving a coherent and stable explanatory structure. On this combined criterion, the strongest tree-based models remain the most convincing, because they repeatedly show that predictive retention and feature-priority retention can coexist rather than trade off sharply against one another.

VIII. WHY TREE-BASED MODELS REMAIN STRONG ON TABULAR DATA

VIII-A. Selective feature usage under nuisance structure

A first explanation is that the strongest tree-based models are more selective in how they use features. The robustness experiments show that, when the input space is made more difficult through missingness or synthetic high-cardinality categorical features, the tree models generally retain both stronger

predictive performance and stronger feature-importance stability. This is clearest under high-cardinality stress. CatBoost shows only a small mean performance delta (-0.0108), while preserving high SHAP overlap and the highest mean Jaccard among the evaluated models. By contrast, the MLP degrades more strongly in predictive terms and exhibits a much larger drop in attribution stability, indicating that the added nuisance structure is not simply ignored but enters the learned representation more materially.

This difference is consistent with the way the model families treat uninformative structure. Tree ensembles perform feature selection implicitly through splitting: a feature that does not improve the objective need not be used at all. High-cardinality nuisance variables therefore create less harm when they fail to produce useful gain. In contrast, neural models often absorb such structure into a dense representation or a larger encoded input, which increases optimization burden even if the added variables carry no genuine signal. The earlier project analyses already described this as a form of distraction by nuisance structure, and the current efficiency results support the same interpretation: under high-cardinality stress, MLP exhibits a very large increase in training time and model size, whereas the strongest tree models remain comparatively stable.

The same logic helps explain why some neural or foundation models did not perform better even when their clean-data baselines were competitive. The issue is not that they never capture useful signal; rather, they appear less able to suppress structurally difficult but uninformative inputs once those inputs enter the representation pipeline. This is one reason why strong clean-data performance does not always translate into equally strong behavior under perturbation.

VIII-B. Attribution stability and reliance on core predictors

A second explanation is that tree-based models preserve a more stable notion of which features matter. This is one of the most important findings of the paper, especially given the emphasis on core predictors, feature types, and target-relevant structure.

Under covariate shift, CatBoost and Random Forest attain the highest mean SHAP stability, with near-perfect mean Spearman correlations and clearly higher Jaccard overlap than MLP or TabNet. The same pattern appears under missingness and high-cardinality stress, where tree models retain roughly eight to nine of their top ten features on average, while neural models preserve fewer. Figure 3 summarizes this relationship directly: the tree-based models cluster closer to the region combining stronger performance retention and higher attribution stability, whereas neural models tend to move downward more strongly in feature-stability space.

This matters because robust prediction and stable explanation are not identical. A model can retain moderate predictive performance while changing which predictors it emphasizes most strongly; conversely, it can preserve the same core features while still suffering performance degradation if the shifted distribution weakens their utility. The strongest tree-based models are more compelling because they tend to preserve both. In that sense, they are not merely more accurate predictors, but also more stable estimators of the underlying explanatory structure. This aligns closely with the feature-centered interpretation developed in the earlier deep drafts: tree models appear to keep focusing on the most informative variables, whereas neural models are more likely to reallocate importance under perturbation or shift.

This also helps explain why current neural and pretrained tabular models did not consistently surpass the tree baselines. Even when competitive in score, they were less consistent in preserving the same feature-priority structure. A model whose dominant predictors drift more under shift is harder to trust as a stable tabular learner, particularly in applications where feature semantics matter and explanations must remain interpretable across deployment conditions.

VIII-C. Feature orientation, redundancy, and irregular target structure

A third explanation concerns the structural character of tabular data itself. The datasets in this study were deliberately chosen to vary in feature composition, target type, cardinality, sample size, and correlation structure. Across these settings, the strongest tree-based models appear better aligned with three recurring properties of tabular data: heterogeneous feature spaces, moderate redundancy, and irregular decision structure.

The earlier project analyses emphasized that tree models preserve the original orientation of the feature space more naturally than neural models. This point should not be stated as a theorem for the present benchmark, but it is consistent with the observed results. Split-based models evaluate individual features or simple partitions directly, whereas neural

models construct hidden representations through learned linear combinations. When the task depends on threshold-like rules, sparse informative columns, or a small set of dominant predictors, the tree bias can be advantageous. This interpretation is supported by the shift analyses, where tree models often retained the same top variables under perturbation, and by the Diamonds comparison, where stronger decorrelation affected neural models more than tree ensembles.

The NeurIPS benchmark paper frames this more generally as a question of tabular inductive bias: tree-based methods appear better suited to preserving feature orientation, handling uninformative variables, and learning irregular functions, whereas neural networks are often biased toward smoother solutions and less well-matched invariances [9]. Our results are consistent with that view. We do not directly replicate their transformation-based experiments, but our missingness, high-cardinality, and shift analyses point in the same direction. The strongest tree models remain stable when the feature space is partially degraded, expanded with nuisance structure, or distributionally displaced. That is difficult to explain by optimization alone; it suggests an underlying match between tree-based decision structure and the statistical form of many tabular problems.

VIII-D. Why neural and pretrained tabular models still fall short

The present results do not imply that neural or pretrained tabular models are uniformly weak. In fact, they remain competitive in selected settings, and the pretrained models reduce part of the gap relative to standard neural baselines on some classification tasks. However, they still fail to match the strongest tree-based models consistently across the combined criteria of clean-data performance, perturbation robustness, attribution stability, and operational efficiency.

For the neural models, the main limitation is not simply insufficient capacity. The pattern across experiments suggests a more specific failure mode: they are more sensitive to feature-space changes, more dependent on favorable hyperparameter configurations, and more likely to redistribute importance across predictors when the input distribution changes. The earlier analyses described this as a tendency to overfit or mis-prioritize features under nuisance perturbation, and the current robustness and shift results support that reading. Even after fair tuning, MLP and TabNet remain less stable than the strongest tree models either in predictive terms, attributional terms, or both.

For the pretrained models, the picture is slightly different. Their main strength is not full dominance over tree ensembles, but reduction of the gap to conventional neural models with less dataset-specific tuning. Even so, they do not consistently surpass the strongest tree-based methods. APT remains weak in overall benchmark ranking, and both APT and TabPFN lack the same breadth of SHAP-based stability evidence available for the core models. More importantly, where predictive data are available, they still do not establish a new robust frontier beyond the boosted tree baselines. Thus, pretraining helps, but does not remove the core tabular inductive-bias problem identified in the literature and reflected in our benchmark.

There is also a practical asymmetry in the comparison. The tree methods used here are ensembles, and ensembling contributes regularization and variance reduction with manageable cost. The neural baselines, by contrast, are single models. In principle, neural ensembles could improve performance, but at substantially higher computational cost. This does not invalidate the comparison, because in practice the relevant comparison is between the strongest widely usable representatives of each paradigm. On that criterion, tree ensembles remain more attractive not only because they perform strongly, but because they do so with less tuning fragility and better robustness-efficiency trade-offs.

IX. DISCUSSION AND CONCLUSION

IX-A. Limitations and Future Prospects

Although the study compares clean performance, robustness stress, and covariate-shift stability across diverse tabular settings, several limitations remain:

- **Benchmark scope:** The benchmark covers different task types and feature structures, but remains moderate in size. The robustness analysis uses four datasets and the covariate-shift analysis three, so the findings should be read as recurring empirical patterns rather than universal claims.
- **Controlled perturbations:** Missingness, high-cardinality stress, and covariate shift were implemented through designed perturbation protocols. These settings isolate specific failure modes, but real deployment shifts may be more complex, temporal, or domain-specific.
- **Interpretability scope:** SHAP-based stability measures whether models preserve similar feature-attribution patterns, but it does not establish causal feature use or complete interpretability. Future work could include local, interaction-level, or alternative attribution analyses.
- **Incomplete model coverage:** Some comparisons remain incomplete: LightGBM could not be fully evaluated in the covariate-shift setting, and SHAP-based stability was unavailable for APT and TabPFN. Therefore, the strongest explanatory-stability claims apply mainly to models with both predictive and SHAP outputs.
- **Future directions:** The next extensions should include probabilistic quality, calibration, uncertainty estimation, and the usefulness of neural or pretrained embeddings for downstream tabular tasks.

IX-B. Conclusion

This paper examined tabular model behavior beyond clean benchmark ranking by combining hyperparameter-optimized evaluation with robustness stress tests and feature-stability analysis. Across these experiments, tree-based models, particularly CatBoost and the other ensembles, remained the most reliable family not only on clean data but also under missingness, high-cardinality nuisance structure, and covariate shift. Their advantage was reflected not only in predictive performance, but also in more stable reliance on core features and better robustness-efficiency trade-offs. These findings suggest that the continued strength of tree-based methods on

typical tabular data is linked to inductive bias rather than benchmark coincidence alone. More broadly, the study argues that tabular models should be evaluated not only by clean test accuracy, but also by robustness, explanatory stability, and practical reliability.

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